# Practice Exam 2

## MTH 207 (Fall 2026) — J. Baggett

**This is practice, not a preview.** The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (134 points against 100) so that it covers more ground, so do not use it to judge timing. The real exam is 55 minutes.

| First Name | Last Name |
|---|---|
| | |

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 8 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

**Geometry, if you need it.** You are not expected to memorize these.

| | | | |
|---|---|---|---|
| Circle | $A = \pi r^2,\ C = 2\pi r$ | Sphere | $V = \frac{4}{3}\pi r^3,\ S = 4\pi r^2$ |
| Square | $A = s^2$ | Cylinder | $V = \pi r^2 h$ |
| Rectangle | $A = \ell w$ | Cone | $V = \frac{1}{3}\pi r^2 h$ |
| Triangle | $A = \frac{1}{2}bh$ | Cube | $V = s^3,\ S = 6s^2$ |
| Right triangle | $a^2 + b^2 = c^2$ | | |

(3pts ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{f(x)g'(x) - f'(x)g(x)}{[g(x)]^2}$.

**Solution:**

**FALSE.** Test $f = x^2$, $g = x$: the quotient is $x$, whose derivative is 1 — but this formula gives $\dfrac{x^2(1) - 2x(x)}{x^2} = -1$. The numerator is backwards; it is $f'g - fg'$, and subtraction does not commute.

(b) If $h(x) = f(g(x))$ with $g(1) = 1$, $g'(1) = 3$ and $f'(1) = 5$, then $h'(1) = 15$.

**Solution:**

**TRUE.** $h'(1) = f'(g(1)) \cdot g'(1) = f'(1) \cdot 3 = 15$.

(c) $\ln(x^2) = 2\ln x$ for every $x \neq 0$.

**Solution:**

**FALSE.** Test $x = -1$: the left side is $\ln 1 = 0$, while the right side is $2\ln(-1)$, which is not defined. The identity that does hold for every $x \neq 0$ is $\ln(x^2) = 2\ln|x|$.

(d) If $f$ is differentiable and $f(-1) = f(1)$, then there is a number $c$ with $|c| < 1$ and $f'(c) = 0$.

**Solution:**

**TRUE.** Rolle's Theorem, and every hypothesis is met: $f$ is differentiable (so also continuous) and takes the same value at both ends, so somewhere between them the tangent must be horizontal.

(6pts) **2.** Evaluate the limit with **justifications**. Simplify until you have a number.

$\lim\limits_{x\to 0} \dfrac{\tan 4x}{\sin 7x}$

**Solution:**

Both $\tan 4x$ and $\sin 7x$ tend to 0, so this is $\frac{0}{0}$ and must be rewritten. Use $\lim\limits_{\theta\to 0} \dfrac{\sin\theta}{\theta} = 1$ by manufacturing the matching denominators:

$$\frac{\tan 4x}{\sin 7x} = \frac{\sin 4x}{\cos 4x} \cdot \frac{1}{\sin 7x} = \underbrace{\frac{\sin 4x}{4x}}_{\to 1} \cdot \underbrace{\frac{7x}{\sin 7x}}_{\to 1} \cdot \frac{4x}{7x} \cdot \frac{1}{\cos 4x}$$

As $x \to 0$ the first two factors tend to 1 and $\cos 4x \to 1$, so

$$\lim_{x\to 0} \frac{\tan 4x}{\sin 7x} = 1 \cdot 1 \cdot \frac{4}{7} \cdot 1 = \boxed{\frac{4}{7}}$$

($6^{\text{pts}}$) **3.** A table of values for $f, g, f'$ and $g'$ is given. If $h(x) = f(g(x))$, find $h'(1)$.

| $x$ | $f(x)$ | $g(x)$ | $f'(x)$ | $g'(x)$ |
|---|---|---|---|---|
| 1 | 3 | 2 | 4 | 6 |
| 2 | 1 | 8 | 5 | 7 |
| 3 | 7 | 2 | 7 | 9 |

**Solution:**
By the Chain Rule, $h'(x) = f'(g(x)) \cdot g'(x)$, so

$$h'(1) = f'(g(1)) \cdot g'(1)$$

From the table $g(1) = 2$, so the outer factor is $f'(2) = 5$; and $g'(1) = 6$.

$$h'(1) = 5 \cdot 6 = \mathbf{30}$$

The usual error is reading $f'(1)$ instead of $f'(g(1))$ — the outer derivative is evaluated at the *inner function's output*, not at $x$.

($6^{\text{pts}}$) **4.** Compute the linearization of $f(x) = x^3$ at the $x$-value 2.

**Solution:**
The linearization at $a$ is $L(x) = f(a) + f'(a)(x - a)$.
Here $a = 2$, $f(2) = 2^3 = 8$, and $f'(x) = 3x^2$ so $f'(2) = 12$. Therefore

$$L(x) = 8 + 12(x - 2) = \mathbf{12x - 16}$$

($6^{\text{pts}}$) **5.** Find the 50th derivative of $y = \cos 2x$.

**Solution:**
Differentiate a few times and watch the cycle:

$$y = \cos 2x \qquad y' = -2\sin 2x$$
$$y'' = -4\cos 2x = -2^2 \cos 2x \qquad y''' = 8\sin 2x$$
$$y^{(4)} = 16\cos 2x = 2^4 \cos 2x$$

Each derivative contributes a factor of 2, and the trig function cycles with period 4. So $y^{(n)} = \pm 2^n \cos 2x$ when $n$ is even.
$50 = 4(12) + 2$, so $y^{(50)}$ sits at the same point of the cycle as $y''$, which carries a minus sign:

$$y^{(50)} = \mathbf{-2^{50}} \cos \mathbf{2x}$$

($9^{\text{pts}}$) **6.** If $y$ depends on $x$ according to $\cos(xy) = 1 + \sin y$, find $\dfrac{dy}{dx}$.

**Solution:**
Differentiate both sides with respect to $x$, remembering $y$ is a function of $x$ (so $xy$ needs the Product Rule inside the Chain Rule):

$$-\sin(xy) \cdot \left( y + x \frac{dy}{dx} \right) = \cos y \cdot \frac{dy}{dx}$$

Expand and collect the $\dfrac{dy}{dx}$ terms on one side:

$$-y \sin(xy) = \cos y \frac{dy}{dx} + x \sin(xy) \frac{dy}{dx} = \frac{dy}{dx} \Big( \cos y + x \sin(xy) \Big)$$

$$\frac{dy}{dx} = \frac{-\mathbf{y} \sin(\mathbf{xy})}{\cos \mathbf{y} + \mathbf{x} \sin(\mathbf{xy})}$$

($7^{\text{pts}}$ ea.) **7.** Derive the following derivative rules:

(a) Derive the derivative rule for $y = \tan x$. (You may use the derivative rules for sine and cosine.) This question is about **regular tangent**, not inverse tangent.

(b) Derive the derivative rule for $y = \sin^{-1} x$. (You may use derivative rules for the trig functions, but not any derivative rules for inverse trig functions.) This question is about **inverse sine**.

(c) Given a fixed number $a > 0$, derive the derivative rule for $y = \log_a(x)$. (You may use the derivative rule for $a^x$.)

($5^{\text{pts}}$ ea.) **8.** Find the derivative of each function below. (Use the derivative rules from the **previous page**!)

(a) $f(x) = 3x^{10} - x^5 + 2x^3 + \sin(x)$

(b) $y = x^{\sin x}$

(c) $f(x) = \Big(\sin x\Big)\Big(\log_4 x\Big) + \ln(x)$

(d) $y = \cos\left(\dfrac{\sin^{-1}(x)}{3x^4}\right) + \sin(5^{\tan x})$

($20^{\text{pts}}$) **9.** A 13 ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec. How fast is the top of the ladder sliding down the wall then? (Be sure to include units.)

**Solution:**

Let $x$ be the distance from the wall to the base and $y$ the height of the top, both functions of time. The ladder is the hypotenuse, so

$$x^2 + y^2 = 13^2 = 169$$

**When** $x = 12$**:** $y = \sqrt{169 - 144} = \sqrt{25} = 5$ ft.

Differentiate the relation with respect to $t$:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \quad \Longrightarrow \quad \frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$$

Substituting $x = 12$, $y = 5$, $\dfrac{dx}{dt} = 5$ ft/sec:

$$\frac{dy}{dt} = -\frac{12}{5}(5) = \mathbf{-12} \text{ ft/sec}$$

The negative sign means $y$ is *decreasing*: the top is sliding **down the wall at** 12 **ft/sec**.

Note the numbers are only substituted *after* differentiating — put $x = 12$ in first and it becomes a constant, with derivative 0.

**THIS IS THE LAST PAGE OF THE EXAM.**

($8^{\text{pts}}$) **10.** Show that the equation $x^3 + e^x = 0$ has exactly one real root.

**Not on Exam 2.** This one uses an argument we take up *after* the exam, so do not let it stall your review this weekend. The worked solution is worth reading once you have done the rest — it is two separate ideas (something exists, and there is only one of it) and that pairing comes back.

**Solution:**
Let $f(x) = x^3 + e^x$. Two things must be shown.
**At least one root (Intermediate Value Theorem).** $f$ is continuous everywhere, being a sum of a polynomial and an exponential. And

$$f(-1) = -1 + e^{-1} \approx -0.63 < 0, \qquad f(0) = 0 + 1 = 1 > 0$$

so 0 lies between $f(-1)$ and $f(0)$, and the IVT gives a $c$ in $(-1, 0)$ with $f(c) = 0$.
**At most one root (the derivative).**

$$f'(x) = 3x^2 + e^x$$

Here $3x^2 \geq 0$ and $e^x > 0$ for every $x$, so $f'(x) > 0$ everywhere and $f$ is **strictly increasing**. A strictly increasing function takes each value at most once, so it cannot have two roots.
Together: **exactly one real root**.

($12^{\text{pts}}$) **11.** Let $f(x) = xe^x - e^x$. Find the absolute maximum point and absolute minimum point on the interval $[-1, 1]$. (Hint: Use the fact that $e > 2$.)

**Solution:**
$f$ is continuous on the closed interval $[-1, 1]$, so the Extreme Value Theorem applies: check the critical numbers and both endpoints.
**Critical numbers.** Factor before differentiating is not needed; use the Product Rule on $xe^x$:

$$f'(x) = (xe^x)' - (e^x)' = (e^x + xe^x) - e^x = xe^x$$

Since $e^x$ is never 0, $f'(x) = 0$ only at $x = 0$, which is in $[-1, 1]$.
**Compare the three values.**

$$f(-1) = -\frac{2}{e}, \qquad f(0) = -1, \qquad f(1) = 0$$

Using the hint $e > 2$: $\dfrac{2}{e} < 1$, so $-\dfrac{2}{e} > -1$. The three values in order are $-1 < -\dfrac{2}{e} < 0$.
**Absolute maximum (1, 0).** **Absolute minimum (0, −1).**

($8^{\text{pts}}$) **12.** Verify that $f(x) = 1/x$ on $[1, 3]$ satisfies the hypotheses of the Mean Value Theorem (clearly state why each hypothesis is satisfied) and find all numbers $c$ that satisfy the conclusion of the Mean Value Theorem.

**Solution:**

**Hypotheses.** $f(x) = \dfrac{1}{x}$ is a rational function, undefined only at $x = 0$. Since $0 \notin [1, 3]$, $f$ is **continuous on** $[1, 3]$ and **differentiable on** $(1, 3)$. Both hypotheses hold.

**Conclusion.** The MVT promises a $c$ in $(1, 3)$ with

$$f'(c) = \frac{f(3) - f(1)}{3 - 1} = \frac{\frac{1}{3} - 1}{2} = \frac{-\frac{2}{3}}{2} = -\frac{1}{3}$$

With $f'(x) = -\dfrac{1}{x^2}$, solve

$$-\frac{1}{c^2} = -\frac{1}{3} \quad \Longrightarrow \quad c^2 = 3 \quad \Longrightarrow \quad c = \pm\sqrt{3}$$

Only $c = \sqrt{\mathbf{3}} \approx 1.73$ lies in $(1, 3)$; $-\sqrt{3}$ is rejected because the MVT only claims a $c$ *inside the interval*.

**THIS IS THE LAST PAGE OF THE PRACTICE EXAM.**